Chapter 14 Student Notes and Labs

Implications for Learning, Work, and Technology: What the Brain's Design Asks of the World We Have Built

Student Notes

What this chapter is for

The earlier chapters described a brain that learns slowly and offline, learns most from its own errors and from other people, runs on a fixed budget of attention and energy, turns rewarded actions into habits, and reshapes itself around what it does. Chapter 14 asks what that design implies for how people learn, work, and use technology, and sorts the practical claims by the strength of the evidence behind them.

If you come from software or machine learning, the chapter is designed to correct three instincts:
1) treating a fact about neurons as a recommendation, when practical claims need tests of how people actually learn and work,
2) trusting the feeling of fluency, when the conditions that feel easiest (rereading, grouped practice, supplied answers) often teach least, and
3) treating automation and AI as pure gains, when tools that do the work can leave their users unpracticed.

Your goal is to be able to explain why some popular brain-based claims fail, which study practices have classroom evidence, why spacing has an optimal gap, why learners misjudge their learning, what the evidence on sleep, attention, and fatigue does and does not show, how offloading and automation affect skill, how AI tutors can help or harm learning, what is known about social media and well-being, and how plastic the adult brain remains.

The babies who learned Mandarin

- Nine-month-olds learned a Mandarin contrast from about five hours with live speakers, as well as Taiwanese infants, but not from the same speakers on video or audio (Kuhl, Tsao, and Liu 2003) (Figure 14.1).
- Toddlers learned verbs in person and over live video chat, not from a recording (Roseberry, Hirsh-Pasek, and Golinkoff 2014); infants learned more from video beside a peer (Lytle, Garcia-Sierra, and Kuhl 2018).
- Neuroscience reaches the classroom through psychology: "a bridge too far" (Bruer 1997). Figure 14.2 maps features to implications.

What does not follow

- Learning styles: believed by over 90 percent of surveyed teachers (Dekker et al. 2012; Newton and Salvi 2020), unsupported by the crossover test (Pashler et al. 2008).

- The Mozart effect: a short spatial boost in 36 students (Rauscher et al. 1993), about 1.4 IQ points on pooling (Chabris 1999), no advantage over other music (Pietschnig et al. 2010).
- No "left-brained" or "right-brained" people (Nielsen et al. 2013).
- Brain training improves the trained tasks but does not transfer (Owen et al. 2010; Simons et al. 2016).

Learning that lasts

- High-utility techniques: practice testing and spaced study; low: rereading and highlighting (Dunlosky et al. 2013).
- Classroom quizzing: 94 versus 81 percent on chapter exams (Roediger et al. 2011).
- Interleaved mathematics homework: 61 versus 38 percent a month later, in 54 classes (Rohrer et al. 2020).
- The best spacing gap is about 20 to 40 percent of a one-week retention interval and 5 to 10 percent of a one-year interval (Cepeda et al. 2008) (Figure 14.3).

The illusion of fluency

- Interleaved paintings: 61 versus 35 percent, yet 78 percent of learners judged blocking as good or better (Kornell and Bjork 2008) (Figure 14.4).
- Failed guesses followed by the answer help memory more than studying the answer (Kornell, Hays, and Bjork 2009); curiosity enhances memory through the midbrain and hippocampus (Gruber et al. 2014).
- Growth mindset: small effects for lower achievers (Yeager et al. 2019), disputed by meta-analyses (Macnamara and Burgoyne 2023; Burnette et al. 2023). Expressive writing before exams (Ramirez and Beilock 2011) failed a large replication (Camerer et al. 2018).

Sleep and the school day

- Puberty delays sleep timing and slows the buildup of sleep pressure (Carskadon et al. 1997; Crowley et al. 2018): a "perfect storm" (Carskadon 2011).
- Seattle's later start: 34 more minutes of sleep and higher grades, without a control group (Dunster et al. 2018).
- Four hours of bright tablet reading suppressed and delayed melatonin (Chang et al. 2015).

Attention is a budget

- Working memory holds about four items (Cowan 2001); task switches cost time even with preparation (Rogers and Monsell 1995; Monsell 2003).
- Media multitasking (Ophir et al. 2009) and smartphone "brain drain" (Ward et al. 2017) claims weakened in replications (Wiradhany and Nieuwenstein 2017; Parry and le Roux 2021; Ruiz Pardo and Minda 2022; Parry 2024).
- Laptop multitasking cost users 11 and nearby peers 17 percentage points (Sana et al. 2013); handwriting's advantage did not replicate (Mueller and Oppenheimer 2014; Morehead et al. 2019; Urry et al. 2021).

- Interruptions: faster work, more stress (Mark et al. 2008); resumed after about 25 minutes (Mark et al. 2005).

How long can a mind work?

- A day of hard cognitive work shifted choices toward easy, quick options, with more glutamate in lateral prefrontal cortex (Wiehler et al. 2022), a correlational finding questioned by Möller (2023).
- Ego depletion did not replicate (Hagger et al. 2016; Vohs et al. 2021); the hungry-judge effect is confounded by case order (Danziger et al. 2011; Weinshall-Margel and Shapard 2011; Glöckner 2016).
- Open-plan offices cut face-to-face interaction by about 70 percent (Bernstein and Turban 2018).

Tools that remember for us

- Plato's *Phaedrus*: writing as a cause of forgetfulness.
- Cognitive offloading (Risko and Gilbert 2016); the "Google Stroop" claim did not replicate (Sparrow et al. 2011; Camerer et al. 2018; Hesselmann 2020).
- Taxi drivers' posterior hippocampi (Maguire et al. 2000), growth in trainees who qualified (Woollett and Maguire 2011); GPS use and weaker spatial memory, modest and small-sample (Dahmani and Bohbot 2020); the hippocampus tracks streets only when people navigate themselves (Javadi et al. 2017).

When the machine does the work

- Bainbridge's ironies of automation (1983); Air France 447 (BEA 2012); endoscopist detection fell after AI exposure, observationally (Budzyń et al. 2025) (Figure 14.5).
- An unrestricted chatbot raised practice scores and lowered exam scores; a hint-giving tutor did not harm them (Bastani et al. 2025) (Figure 14.6). A guided physics tutor doubled immediate learning gains (Kestin et al. 2025). EEG evidence is preliminary (Kosmyna et al. 2025, preprint).

Designed to be hard to put down

- Posting follows reward learning (Lindström et al. 2021); app design features (Montag et al. 2019).
- Technology use explains at most 0.4 percent of adolescent well-being (Orben and Przybylski 2019; reply by Twenge et al. 2020); deactivation improved well-being by about 0.09 SD (Allcott et al. 2020); Facebook's arrival worsened college mental health (Braghieri et al. 2022); limiting use reduced loneliness (Hunt et al. 2018); the dispute between Haidt (2024) and Odgers (2024).

Learning across a lifetime

- Juggling changed gray matter in young and older adults (Draganski et al. 2004; Boyke et al. 2008); learning photography improved older adults' memory (Park et al. 2014).
- Exercise and the hippocampus: growth without a memory difference (Erickson et al. 2011; Coen et al. 2011); no robust effect in a meta-analysis (Balbim et al. 2024).

- Figure 14.7 sorts the chapter's claims by evidence.

Implications for artificial intelligence

- Tutors should make learners do the work: retrieve, guess, and return after a gap.
- Automation should keep its users practiced.
- Systems should help people judge them: higher confidence in AI goes with less critical thinking (Lee et al. 2025).
- Attention is the scarcest resource a tool consumes.

Practical summary

- Retrieve, space, and mix; sleep.
- Distrust fluency: what feels easy often teaches least.
- Divide attention as little as possible; limit long stretches of hard work.
- Use tools that make you do the work, and keep practicing what machines do for you.
- Ask what kind of evidence stands behind a "brain-based" claim.

Labs

Lab Notes: how to succeed in these labs

These labs turn the chapter's claims into short calculations and judgments: how the spacing model chooses a gap, how large the classroom effects are, how to read an effect size, how automation changes a skill, and how to weigh a study. The arithmetic is light; the point is to see where the numbers come from and how much weight they can bear.

General rules:
1) Write every intermediate value, rounded to two decimal places unless told otherwise.
2) State every assumption you add, and say which way it would move the answer.
3) End with an interpretation: what does the calculation show, and what does it leave out?

What to hand in for every lab

- A table of values.
- A diagram or sketch where requested (a photo of a hand drawing is fine).
- A short interpretation (6-12 sentences).

Lab 14.1 - Choosing a Spacing Gap

Purpose

To see how the illustrative spacing model of Figure 14.3 trades the benefit of a harder review against the risk of forgetting.

Given values

Retrievability after t days: R = 1 / (1 + t / (9 S)), with stability S in days. After the first session, S0 = 2.
At a review after a gap g: if the item is recalled (probability R), stability becomes S2 = S0 × (1 + 4 × (e^(4(1 − R)) − 1)); if not, it stays at S0.
The final test comes 35 days after the review.

Procedure

1) For gaps of 0, 2, 7, and 30 days, compute R at the review. (Check: 1.00, 0.90, 0.72, 0.375.)
2) Compute S2 for a successful review at each gap. (Check: 2.0, 5.9, 18.5, 91.5.)
3) Compute retention at the test if the review succeeded, R(35, S2). (Check: 0.34, 0.60, 0.83, 0.96.)
4) Compute the expected retention, R × R(35, S2) + (1 − R) × R(35, S0). (Check: 0.34, 0.58, 0.69, 0.57.) Which gap is best among these?
5) Use Cepeda's rule of thumb to choose a review gap for an exam one year away (5 to 10 percent) and one week away (20 to 40 percent). (Check: about 18 to 36 days; about 1.4 to 2.8 days.)
6) Write 6-10 sentences: why does the expected retention fall at the longest gap, and what in a real course might make the model's answer wrong?

Lab 14.2 - How Big Are the Classroom Effects?

Purpose

To express classroom results in absolute and relative terms.

Given values

Roediger et al. (2011): chapter exams 94 percent (quizzed) and 81 percent (not quizzed); semester exam 79 and 67 percent.
Rohrer et al. (2020): 61 percent (interleaved) and 38 percent (blocked) a month later.
Kornell and Bjork (2008): 78 percent of participants did better with interleaving; 78 percent judged blocking as good or better.

Procedure

1) Compute the absolute and relative gains from quizzing on the chapter and semester exams. (Check: 13 points, 16.0 percent; 12 points, 17.9 percent.)
2) Compute the absolute gain from interleaving and the ratio of the two scores. (Check: 23 points; 1.61.)
3) Knowing nothing about how the two 78 percent groups in Kornell and Bjork overlap, what is the smallest possible share of participants who both did better with interleaving and judged blocking as good or better? (Hint: 78 + 78 − 100. Check: at least 56 percent.)
4) Write 6-10 sentences: why might classroom effects be smaller than laboratory ones, and why might students keep using the weaker techniques?

Lab 14.3 - Reading an Effect Size

Purpose

To translate standardized effect sizes into something intuitive.

Given values

For an effect of d standard deviations, the average treated person scores higher than a fraction $\Phi(d)$ of the control group, where Φ is the standard normal cumulative distribution.

Procedure

1) Compute $\Phi(d)$ for d = 0.04 (ego depletion), 0.09 (Facebook deactivation and well-being), 0.11 (growth mindset for lower achievers), 0.83 (interleaved mathematics), and 1.0. (Check: 51.6, 53.6, 54.4, 79.7, 84.1 percent.)
2) Technology use explained at most 0.4 percent of the variation in well-being. Convert to a correlation r = $\sqrt{0.004}$. (Check: about 0.063.)
3) Rank the five effects, and say for each whether it is large enough to matter for an individual, for a population, or for neither.
4) Write 6-10 sentences: why can a small average effect still matter for a large population, or be large for a subgroup?

Lab 14.4 - The Cost of Switching

Purpose

To estimate how task switches and interruptions add up over a working day.

Given values (hypothetical, for the exercise)

Each switch between tasks costs 1 second of slowed work. A worker switches 20 times an hour for 8 hours. In addition, 4 interruptions a day each take the worker away from the main task, which is resumed after 25 minutes on average.

Procedure

1) Compute the time lost to switch costs in a day. (Check: 160 seconds, under 3 minutes.)
2) Compute the time the main task is set aside because of interruptions. (Check: 100 minutes.)
3) Which number is larger, and which does the chapter's evidence suggest matters more: the time lost, or the stress and fragmentation Mark and her colleagues measured?
4) Redesign the day to halve the interruptions without reducing the work done. Describe two concrete changes.
5) Write 6-10 sentences: why are the dramatic claims about multitasking weaker than the switch cost itself, and what does that imply for workplace policy?

Lab 14.5 - Skill Under Automation

Purpose

To work out the equilibrium of the automation model of Figure 14.5.

Given values

Each day, skill s rises by g × (1 − s), where g = 0.1 × (1 − f) and f is the fraction of tasks automated, and then falls by 0.4 percent of its value.
At equilibrium, $s^* \approx g / (g + 0.004)$.

Procedure

1) Compute s^* for f = 0, 0.5, 0.9, and 0.99. (Check: 0.96, 0.93, 0.71, 0.20.)
2) Plot s^* against f. Where does the curve bend most steeply?
3) A hospital wants to keep endoscopists' skill at 0.9 or above while using AI. What fraction of cases must they do without it, in this model? (Hint: solve $s^* = 0.9$ for g, then for f. Check: g = 0.036, so f = 0.64; at least 36 percent of cases by hand.)
4) Write 6-10 sentences relating the result to Bainbridge's final irony and to the Air France 447 report.

Lab 14.6 - Weighing a Study

Purpose

To practice judging how much a single study can bear.

Procedure

1) For each of Kuhl et al. (2003), Dunster et al. (2018), Budzyń et al. (2025), and Bastani et al. (2025), record: the sample size, whether assignment was random, whether there was a control group, how long after the intervention the outcome was measured, and whether the result has been replicated.
2) Give each study a grade from A (strong) to D (weak) for its main claim, and justify the grade in two sentences.
3) For one study, design a follow-up that would answer its main weakness.
4) Write 6-10 sentences: which of the four claims would you act on now, and why?

Lab 14.7 - The Evidence Ledger (concept map)

Purpose

To connect the chapter's practical claims to the brain mechanisms of earlier chapters and to the strength of the evidence.

Procedure

1) Copy the claims in Figure 14.7.

2) For each claim, name the brain mechanism it rests on and the chapter where that mechanism was described (use Figure 14.2).
3) Mark claims in which the mechanism is well established but the practical claim failed (for example, plasticity and brain training), and explain the gap.
4) Add two claims you have encountered outside this book, and place them in the ledger with your reasons.
5) Write 8-12 sentences: what kind of evidence should a school or company require before adopting a “brain-based” practice?